Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694, 089 632 5829 · www.solaress.org

BEST-PERFORMANCE PV DESIGN

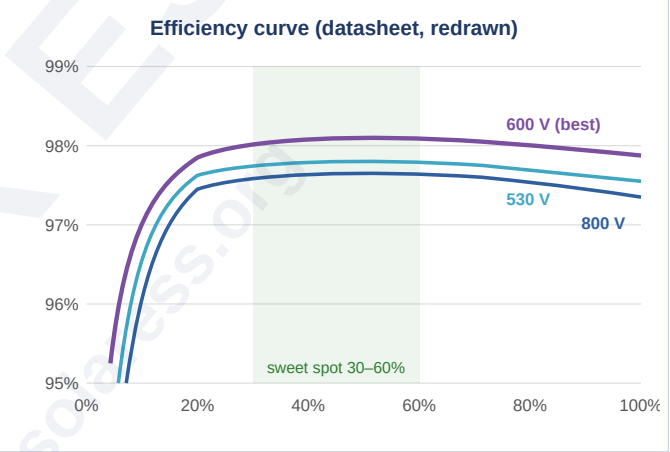
Huawei SUN2000-50KTL-M3 with LONGi Hi-MO X10 LR8-66HVD-650M Anti-Glare (650 Wp, bifacial), matched to the inverter's MPPT sweet spot

Result. The SUN2000-50KTL-M3 is most efficient at an MPPT voltage of about **600 V** and at 30–60% load. With this panel, **15–16 panels per string** keep the operating voltage at 580–620 V for most of the day. Two designs come out best:

- **Best performance: 8 strings × 15 panels = 120 panels, 78.0 kWp (DC/AC 1.56)** — about **111,000 kWh/year**, total current-limit and clipping loss under 1%.
- **Highest efficiency: 6 strings × 16 panels = 96 panels, 62.4 kWp (DC/AC 1.25)** — about **89,300 kWh/year**, the highest yield per kWp, no clipping.

1. Where the M3 works best

SUN2000-50KTL-M3	Value
Max. / European efficiency	98.5% / 98.0%
Rated input voltage	**600 V**
MPPT range / start voltage	200–1,000 V / 200 V
Max. input voltage	1,100 V
MPPTs / inputs	4 / 8 (2 per MPPT)
Max. current per input	20 A
Max. current per MPPT	30 A (all four)
Max. short-circuit current per MPPT	40 A
Rated / max. AC power	50 kW / 55 kW



Read from the datasheet curve: at 600 V the M3 peaks at about 98.3%. At 530 V and 800 V it runs about 0.4–0.6% lower. Efficiency is flat between 30% and 60% load and drops below 15% load. The design goal is therefore an operating MPPT voltage close to 600 V, and an array large enough that the inverter spends most daylight hours above 30% load.

2. Matching the panel to the MPPT point

Panels per string	Voc at 0 °C (limit 1,100 V)	Vmp at 70 °C (hot noon)	Vmp at 45 °C (typical operation)	Vmp at 25 °C	Daytime average Vmp	Fit to 600 V
14	731 V	509 V	545 V	574 V	545 V	**Slightly low**
15	**783 V**	**546 V**	**584 V**	**615 V**	**583 V**	**Excellent**
16	**835 V**	**582 V**	**623 V**	**656 V**	**622 V**	**Excellent**
18	940 V	655 V	700 V	738 V	700 V	**Good (−0.2%)**
20	1,044 V	727 V	778 V	820 V	778 V	**Far from 600 V; little voltage margin**

Panel data: Voc 49.72 V, Vmp 40.98 V, Imp 15.86 A, Isc 16.54 A; Voc −0.20 %/°C, Vmp −0.25 %/°C. "Daytime average Vmp" is weighted by irradiance over a modelled Chiang Rai year. 15- and 16-panel strings sit on both sides of 600 V, so the inverter runs on its best efficiency curve most of the time.

3. Current check (M3 limits are tighter than the MC0)

Item	Limit	1 string	2 strings	Comment
Operating current per MPPT (Imp, +10% rear gain at 1,000 W/m²)	30 A	**17.4 A**	**34.9 A**	Two strings exceed 30 A only above about 860 W/m², a few midday hours on clear days: estimated loss 0.3–0.5% a year
Short-circuit current per MPPT, roof / normal ground	40 A	**18.6 A**	**37.2 A**	OK
Short-circuit current per MPPT, carport / light ground	40 A	**20.3 A**	**40.6 A**	Two strings not allowed: use one string per MPPT
Current per input	20 A	**15.9–17.4 A**	—	OK in normal operation

4. Design options compared

Layout (strings on MPPT 1/2/3/4)	Panels	kWp	DC/AC	Avg. Vmp	Est. kWh / year	kWh per kWp	Current-limit loss	AC clipping
4 × 20 (1/1/1/1)	80	52.0	1.04	778 V	74,300	1,430	0%	0%
6 × 16 (1/2/2/1) — highest efficiency	**96**	**62.4**	**1.25**	**622 V**	**89,300**	**1,431**	**0.3%**	**0%**
6 × 18 (1/2/2/1)	108	70.2	1.40	700 V	100,200	1,427	0.3%	0%
8 × 15 (2/2/2/2) — best performance	**120**	**78.0**	**1.56**	**583 V**	**111,000**	**1,423**	**0.5%**	**0.3%**
8 × 16 (2/2/2/2)	128	83.2	1.66	622 V	117,200	1,409	0.5%	1.3%
8 × 18 (2/2/2/2)	144	93.6	1.87	700 V	127,900	1,367	0.5%	4.0%

Estimated with a simplified model of Chiang Rai sunshine (about 1,430 kWh/kWp/year), roof or normal-ground mounting (+10% rear gain), the M3's 30 A MPPT limit, the voltage and part-load shape of the datasheet efficiency curve, and the 55 kW AC limit. Not based on site weather data; use the figures to compare designs. The 4 × 20 layout has no current loss but runs far from 600 V and lightly loads the inverter; its yield per kWp only matches because the model's voltage penalty is small.

5. Recommended layouts

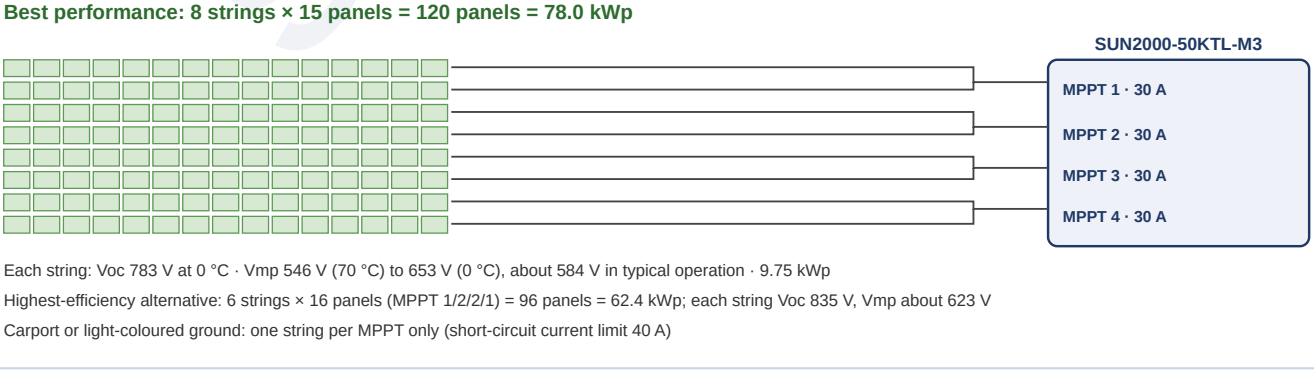


Figure 2 — Best-performance string layout. Each pair of strings joins at the inverter's own two inputs per MPPT.

6. Recommendation

- **Best performance: 8 strings × 15 panels = 120 panels (78.0 kWp).** The MPPT voltage sits right at the inverter's 600 V sweet spot, the inverter spends most of the day in its 30–60% high-efficiency band, and current-limit plus clipping losses stay under 1%.
- **Highest efficiency per panel: 6 strings × 16 panels = 96 panels (62.4 kWp).** Choose this when space or budget is limited; it also leaves two inputs free (MPPT 1 and 4) to add two more 16-panel strings later.
- Avoid 20-panel strings on the M3: they run far above 600 V and leave little margin to the 1,100 V limit.
- On a carport or light-coloured ground, the M3 allows only one string per MPPT (short-circuit current over 40 A with two). Use the MC0 there instead, which allows 44 A on MPPT 2 and 3.
- Strings on the same MPPT must have the same panel count, orientation and tilt. Ask our supplier to confirm that Huawei's warranty covers DC/AC 1.56; for a grid-connected system, also check PEA's requirements.

Sources: Huawei SUN2000-50KTL-M3 datasheet (technical specification and efficiency curve); LONGi LR8-66HVD-650M specifications. Energy figures are model estimates; confirm the final design against the latest datasheets and site conditions.

SOLAR ESS
www.solaress.org